Ali *et al. Smart Learning Environments* (2025) 12:3
https://doi.org/10.1186/s40561-024-00357-y

Smart Learning Environments

RESEARCH Open Access

Investigating continuous intention to use metaverse in higher education institutions: a dual-staged structural equation modeling-artificial neural network approach

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Abstract

The current study explores metaverse adoption among higher education institutions (HEIs) in the light of a theoretical framework to empower future perspectives of the metaverse as a learning platform. Even though this technology was just recently introduced to the higher education sector, very few attempts have been made to evaluate its impact. The purpose of this research is to analyze the elements that influence the continuous intention (CI) to utilize the metaverse technology in learning. The technology acceptance model (TAM) and the self-determination theory (SDT) are both included in this study. A questionnaire was developed and distributed to students attending private universities in order to obtain the data that was needed for the proposed model. Using a hybrid approach that consists of partial least squares structural equation modeling (PLS-SEM) and an artificial neural network (ANN) model, which combines a linear PLS model with compensation and a nonlinear ANN model without compensation, the effect of CI on using the metaverse as a learning platform is investigated. This approach was chosen because it contains both of these types of models. When it comes to explaining the use of metaverse technology among students attending higher education institutions in Egypt, the research findings suggested that autonomy and perceived usefulness (PU) are major determinants. Nevertheless, the continuing intention was unaffected by the perceived ease of use (PEOU) of the product. Furthermore, according to the data provided by the ANN model, the most significant predictors are relatedness, PEOU, autonomy, and PU. It has been determined that the results obtained from the PLS-SEM and ANN modes are identical. Additionally, both theoretical and practical implications are discussed in this article.

Keywords: Continuous intention, Metaverse, TAM, SDT, Neural network, PLS-SEM, ANN

Introduction

The context of higher education is constantly shifting as a result of the proliferation of new technologies and the spread of globalization. The majority of educational institutions, regardless of their size or shape, battle to compete by enhancing their technological

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skills (Khalid et al., 2018). Recently, there have been many changes to the higher education sector scene within and after the pandemic. Many HEIs globally, including Egypt, have transitioned to virtual learning. The challenge that COVID-19 poses to the world's education sector is profound (He et al., 2022). Many education systems have recently been forced to take the learning process and other operations remotely (Comelli et al., 2021). This path results in innovative strategies to cross digital gaps and overwhelm other issues (Salloum et al., 2023).

Metaverse refers to the online virtual world (Metwally et al., 2024). According to Chua and Yu (2024), this new frontier has the potential to be a game-changer for businesses all over the world. It is a magical zone where real life and virtual reality mix to produce an experience that is immersive and entertaining for numerous users (Chen et al., 2023a, 2023b; Mystakidis, 2022). Researchers have previously associated the term "metaverse" with technologies such as augmented reality (AR), extended reality (XR), virtual reality (VR), and mixed reality (MR). These technologies make use of a three-dimensional (3D) virtual world to provide users with an experience that is both immersive and collaborative (Chen et al., 2023a, 2023b; Tan et al., 2023). During the years preceding the COVID-19 pandemic, the foundation of the pandemic was established in the virtual world as a means of overcoming constraints brought about by individual demands and responsibilities involved in extracurricular activities. Additionally, the influence of online education may have been of a higher quality; nevertheless, it was hampered by a number of issues, such as inadequate technological conditions, infrastructure, software applications, and other issues (Lu et al., 2024; Park & Kim, 2022).

Many issues that need to be addressed immediately in online teaching have surfaced. Lack of context, engagement, and participation in online instruction must be addressed quickly (Kaddoura & Al Husseiny, 2023; Zhang et al., 2022). Recent advances in metaverse technology have greatly benefited online education (Wang & Shin, 2022). Because Generation Z (born after 1995) is so different from their predecessors, they are particularly open to the (Park & Kim, 2022). Metaverse can be utilized to teach in numerous circumstances (Kim et al., 2022). Many firms, including Meta, Roblox, Zepeto, and others, are developing metaverse classroom tools. Curriculum creation includes the metaverse, making it more engaging for pupils. While metaverse technology has immense potential in the classroom, its adoption is still young. Free mobile platforms are worse for education (Zhang et al., 2022). Thus, users might be drawn to the metaverse's education ecosystem by studying how the application's platform affects their objectives. It may also fill the requirement for certain instructional technology (Xu & Impagliazzo, 2024).

As a result of the metaverse system's relative novelty, a limited amount of research has been conducted on the acceptability and deployment of the system by users in industrialized countries. According to Tlili et al. (2022b), Metaverse looks to be beneficial to its users in terms of facilitating better classroom administration. As a consequence, there is a significant possibility that this study will shed light on the elements that influence the adoption of metaverse technology by information technology. In this study, we hope to contribute to the existing body of literature on educational technology application platforms in formal educational environments in three different ways. This will allow us to fill the gaps that have been identified. Initially, previous research has focused on

the intention to implement various platforms for educational technology platforms. The majority of research focuses on e-learning (Nayak et al., 2022), mobile learning (Alowayr, 2022), educational mobile games (Tlili et al., 2024), learning management (Al-Nuaimi & Al-Emran, 2021; Tlili et al., 2022a), augmented reality and virtual reality technologies (Jang et al., 2021; Papakostas et al., 2022), and social media services (Al-Rahmi et al., 2021; Yu, 2020). However, Metaverse educational technology differs from traditional educational technology in many ways, including pedagogical aims, intended learners, necessary technological infrastructure, course content, and accessibility to technical (Park & Kang, 2021; Tlili et al., 2023). Therefore, further research into the factors that encourage users to participate in the metaverse is consistently required.

Second, artificial neural networks (ANN) are used in conjunction with partial least squares structural equation modeling (PLS-SEM) to rank the normalized importance of the predictors in order to validate the results of PLS-SEM (Lee et al., 2023; Tan et al., 2014). As a result, this research not only contributes significantly to our theoretical understanding of the uses of virtual learning in educational contexts, but it also offers practitioners vital insights. Thirdly, this research makes a contribution to the existing body of knowledge by combining the three theories of self-determination theory (SDT) and the technology acceptance model (TAM). It provides a comprehensive explanation of the elements that influence the continual intention of Egyptian students to utilize the metaverse. A lack of rigorous theoretical knowledge and empirical research studying the elements of ongoing intention to employ metaverse in learning is the specific problem that this study addresses. In conclusion, this study addresses the specific problem discussed in the previous sentence. As a result, the primary research question that is motivating this study is as follows: What are the key factors and the most crucial ones that influence students' continuing intention to use the metaverse in learning?

Literature review

The metaverse's classroom benefits have piqued academic curiosity. The metaverse definition of education is the main subject of certain studies. Metaverse technology creates a novel learning environment (Morganti & Bartolomei, 2024; Wang & Shin, 2022). The distinct advantages of the educational Metaverse have aroused the curiosity of scholars (Wang et al., 2024). Initially, there is a body of work that primarily addresses the educational context when discussing the Metaverse. Metaverse technology has created a new kind of learning environment that combines four different kinds of augmented reality, virtual reality, life logging, and a mirror world (Salloum et al., 2023). Additionally, other studies shed light on the features of the Metaverse as they pertain to educational settings. The 5Cs—currency, continuity, canon, creator, and connectivity—were put forth by Go et al. (2021) as characteristics of the Metaverse. Compared to traditional classroom instruction, screen-based remote instruction, and instruction in the Metaverse, the former three have many limitations in terms of time and space, while the latter two have many advantages, such as strong interaction, virtual identity, immersive experience, open and accessible creation, and thorough evaluation of teaching (Zhang et al., 2022).

Thirdly, various studies have addressed the universe's most important educational resources and technologies (Büyüközkan & Mukul, 2024). Tlili et al. (2022b) classified a wide range of technologies and approaches, including immersive modeling and

simulation, game applications, artificial intelligence (AI), education, mobile, sensors, and wearable devices. In addition, cloud computing, artistic intelligence, blockchain, 5G, graphics processing units (GPUs), VR, AR, photography 3D engines, and extended reality (XR) were highlighted by Contreras et al. (2022) as having a supporting role in the enhancement of the educational Metaverse. Fourth, a different angle describes the benefits and ways in which Metaverse alters the educational setting. In their seven-pronged analysis of the Metaverse's beneficial effects on education, Lin et al. (2022) highlight its capacity to improve content visualization, boost learning efficiency, release previously limited educational resources, and reduce educational costs and risks. In addition, pupils now have more freedom and opportunity to think creatively because of the educational application of Metaverse software (Kye et al., 2021).

Fifthly, there was an examination of the scholastic situation and an evaluation of the universe's impact. Some of the many potential case-based educational applications described by Hu et al. (2022) include an open university, an immersive curriculum, a learning tool, a twin campus, and a virtual school, among others. Wang and Shin (2022) emphasize the benefits of the Metaverse in terms of simplicity, interactivity, immersion, usability, and interest in educational applications through user analysis of the usage effect of the spatial system learning platform. Investigating how virtual teaming affects education was the focus of Jovanović and Milosavljević (2022). Finally, the Metaverse has raised concerns among certain academics who worry it would disrupt traditional classroom practices. Wang and Shin (2022) argue that there are ethical and privacy concerns with implementing Metaverse technology into the classroom. Problems with implementing the technology, insufficient computer resources, and potential risks to students' physical and mental health are also factors to be taken into account (Hu et al., 2022). Finally, the metaverse has some academics worried that it will ruin classroom instruction. Wang and Shin (2022) argue that there are ethical and privacy concerns with incorporating metaverse technology into the classroom. When it comes to utilizing the metaverse for educational purposes, Kaddoura and Al Husseiny (2023) highlighted important obstacles, ethical concerns, and possible dangers. Problems with implementing the technology, insufficient computer resources, and potential risks to students' physical and mental health are also factors to be taken into account (Nguyen et al., 2024). Nevertheless, there is a dearth of scholarship that investigates the factors that influence students' long-term intentions to use Metaverse for learning. Therefore, this research sets out to address such a knowledge gap.

Furthermore, prior research on educational technology has made use of the TAM or a combination of the TAM with numerous theories (such as ECM, TPB, IDT, TTF, or UTAUT) (Al-Rahmi et al., 2021; Guo et al., 2022; Huang et al., 2020). Researchers have found that TAM and SDT models align well with data on the uptake of technology for online education (Racero et al., 2020). Furthermore, the PLS-SEM is used by the vast majority of empirical studies investigating the goals of educational technology. However, a few studies have discovered the integration of PLS-SEM and ANN. ANN is typically used to handle nonlinearity and multicollinearity well and provide highly accurate predictions (Abdullah et al., 2022). Finally, according to the literature, the integration of

TAM with SDT to examine metaverse in learning is unknown. This study contributes by predicting students' continuing intention to use metaverse through TAM and SDT.

Theoretical background

Technology acceptance model (TAM)

According to Legris et al. (2003), the TAM theory was developed by Davis (1989) and investigates how the introduction of a new system or elements of technology can influence the internal attitudes, beliefs, and intentions of users. PEOU and PU are explanations for users' technical adoption of new systems and technologies, as stated by Legris et al. (2003). According to Davis (1989), PEOU is an indication that technology would be an easy task, and PU is a performance booster. According to Tan et al. (2014), PEOU/PU has an effect on both attitudes and behaviors. In doing so, they establish a connection between future technology use and external circumstances. Nikou and Economides (2017) state that the components and linkages of TAM are not capable of accurately predicting the performance of students from varied backgrounds. When it comes to technology, some pupils are natural users. Fear of technology is a barrier that hinders some people from making use of their technology. Lai (2020) has revealed that PEOU and PU provided an explanation for the acceptance of technology. Both Davis (1989) and Lin et al. (2011) demonstrated that these elements are responsible for the formation of behavioral intentions. PU and PEOU outline the goals of online education in terms of how easy it is to use and how simple it is to understand. They play important roles in the procedure of technology acceptability (Alsabawy et al., 2016; Chang et al., 2017). According to Liesa-Orús et al. (2023), the PU increases in proportion to the PEOU with a higher value. It was discovered by Davis (1989) that even if people find it difficult to use technology, they may still choose to do so because it is helpful and makes their work easier. External TAM variables provide researchers with assistance in predicting the uptake of technologies. In addition to this, it offers explanations for selecting the proper technology, which in turn prompts researchers and specialists to take preventative measures (Davis et al., 2023).

Self-determination theory (SDT)

SDT holds that humans are dynamic and naturally evolve psychologically (Sheeran et al., 2020). Humans naturally seek adventure, new experiences, and education. Internalization also shows it, according to Ryan (2009). By addressing basic psychological needs, SDT creates the framework for development, honesty, and wellness. It recognizes competence, independence, and interdependence (Howard et al., 2020). SDT is particularly relevant to technology adoption because it focuses on intrinsic motivation factors (autonomy, competence, and relatedness) that are critical for ensuring people adopt, engage with, and continue using technologies. The theory provides a deep psychological understanding that can be applied to the design of tech products, improving their adoption rates (Ryan, 2009).

Research model and hypotheses development

TAM and SDT are both incorporated into the model that is being developed by the current research. The continuing intention to use the metaverse platform is impacted by both PU and PEOU, according to the findings of TAM. Within the framework of SDT, it has been presented as a proposition that autonomy, relatedness, and competence have a major impact on PU and ease of use. Furthermore, the research model that has been proposed implies that the factor of autonomy plays a role in the continuing intention to use the metaverse platform. The research model that has been proposed is shown in Fig. 1.

Autonomy

In autonomy, self-regulation is the key, not external interference. Being independent means taking command of one's life. It means learners have to take charge of their learning. Students choose to study (Nikou & Economides, 2017). Students feel empowered to learn and achieve their goals (Adams & Khojasteh, 2018). Student satisfaction affects autonomy. Autonomy-based motivation boosts happiness (Joo et al., 2018). Racero et al. (2020), Nikou and Economides (2017), and Rezvani et al. (2017) demonstrated favorable associations between perceived autonomy, PU, and PEOU in ICT environments. Cortez et al. (2024), Soliman et al. (2024), and Liaw et al. (2010) found a strong link between autonomy and technological acceptability. These findings led to the following hypotheses:

H1: Autonomy has a positive effect on PU.
H2: Autonomy has a positive effect on PEOU.
H3: Autonomy has a positive effect on continuous intention.

Relatedness

Becoming "related" signifies wanting to join a larger fraternity. "Relatedness" in the classroom allows students to collaborate and interact (Sergis et al., 2018). Socializing and

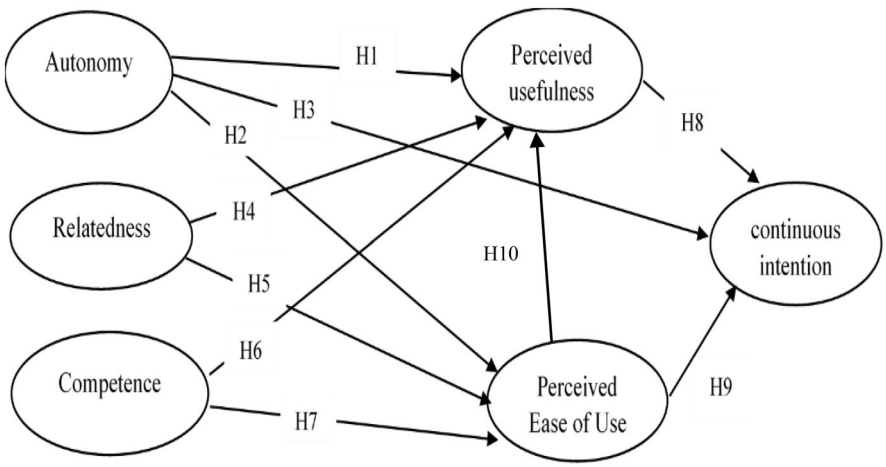


Fig. 1 The proposed research framework

making friends can benefit pupils, according to SDT (Adams & Khojasteh, 2018). Individuals may experience a greater sense of ease when discussing knowledge due to the relatedness. According to Cortez et al. (2024), Soliman et al. (2024), and Racero et al. (2020), relatedness predicted PEOU and PU. Therefore, the following hypotheses have been proposed:

H3: Relatedness positively affects PU.
H4: Relatedness positively affects PEOU.

Competence

Competence helps people achieve their goals and excel. A perception of value is achieved (Ryan & Deci, 2017). Previous research on education competence linked PU to PEOU (Jeno et al., 2017; Roca & Gagné, 2008). Students must master online learning to succeed in class (Niemiec & Ryan, 2009). The following hypotheses were derived from these results:

H5 Competence has a positive effect on PU.
H6 Competence has a positive effect on PEOU.

Perceived usefulness and perceived ease of use

An individual's level of confidence in the convenience and security of a system is measured by the PEOU. In contrast, Perceived Usefulness (PU) is the belief that a system would improve job performance (Davis, 1989). Following studies of Soliman et al. (2024), Racero et al. (2020), Diop et al. (2019), and Al-Rahmi et al. (2021), PU and PEOU influence technological adoption. The following hypotheses were derived from these results:

H8: PEOU positively affects students' continuous intention to use the metaverse platform.
H9: PU positively affects students' continuous intention to use the metaverse platform.
H10: PEOU positively affects students' PU when using the metaverse platform.

Methods

Population and sample

In the course of the academic year 2022/2023, this study was carried out by means of a survey, which included a quantitative research approach for the purpose of data collection. Students from private universities who were members of an information technology college were the subjects of this study. Virtual environment platforms, such as augmented reality (AR), virtual reality (VR), extended reality (XR), and mixed reality (MR), among others, were already familiar to them just by virtue of their existence. It is likely that they learnt about the metaverse from their peers or from participating in social media. To further familiarize the intended respondents with the overall structure of the metaverse learning environment, we also included a link to a YouTube film that

was titled "Discover a metaverse built for education" in the survey. The chosen institution gave its consent for the study to proceed with ethical considerations. The involvement of the responders was entirely non-essential. One of the instruments that was utilized was a web-based survey that was constructed using Google Forms. Purposive sampling was the method that was utilized in order to pick the people who were the subjects of consideration. In order to ascertain the minimal sample size that was necessary, the G*Power tool (Faul et al., 2009) was deployed. It is important to note that the G*Power parameters consist of the following values: 0.15 represents a medium effect size, 0.05 represents the error type (α), 0.80 represents the power, and five represents the number of predictors.

In light of this, it was determined that 92 cases constitute the bare minimum sample size that is required. All of the information, including the purpose of this study and the link to the poll in question, was distributed to the students using WhatsApp groups. A total of 145 students participated in the online survey, and the questions require responses in order to prevent any data from being overlooked. Students who identify as male makeup around 58% of the total sample. A total of 42% of the sample population is comprised of females. These individuals range in age from 18 to 23 years old. It is possible that the quality of the data collected and the validity of the findings of the study might be improved if the participants voluntarily donated their time and effort. Since students participate in the metaverse technology study of their own volition, the input they provide is, therefore, more reliable. A structural equation modeling approach, utilizing the SmartPLS 4.0 software, was utilized in order to investigate the direct and indirect interactions that occurred between the variables under investigation. Using IBM SPSS 26, we also performed an analysis of descriptive data in addition to the ANN technique.

Instruments

The first component of the survey is dedicated to collecting the demographic information of the people who participated in the survey. As part of the second portion, the constructs of the conceptual model were evaluated. These constructs included PEOU, PU, autonomy, relatedness and competence, and continuous intention. A "7-point Likert scale" was utilized in order to analyze their responses. The evaluation of autonomy and competence consisted of four items for each of the two categories. The items were sourced from Nikou and Economides (2017) and Lee et al. (2015). For example, a sample item of autonomy is: “I feel a sense of choice and freedom using the Metaverse platform.” Meanwhile, a sample for competence is: “I have stronger capability than other users, thanks to the Metaverse platform.” Relatedness was evaluated by employing a set of four items taken from Lee et al. (2015) and Sørebø et al. (2009). A sample item is: “Metaverse platform gives me more chances to interact with others.” Further, PU and PEOU were assessed using four and three items, respectively. These items were adopted by Venkatesh et al. (2003) and Nikou and Economides (2017). For example, a sample item for PU is: “Using the Metaverse platform increases my productivity.”, while a sample item for PEOU is: “I find the Metaverse platform easy to use.” Also, three items of continuous intention were adopted from Bhattacherjee (2001). A sample item is: “I intend to continue using the Metaverse platform rather than other alternative means.” Finally,

the marker variable was evaluated by employing a set of three items taken from Lin et al. (2015). A sample item is: "Once I have concluded, I'm not likely to change my mind."

Data analysis

In this study, the proposed research model was analyzed with the use of the SmartPLS 4.0 software tool, which was based on the Partial Least Squares (PLS) analytic technique (Ringle et al., 2022). Assessment of the measurement model, which encompasses the validity and reliability of the measurements, and assessment of the structural model, which includes verifying the hypothesized relationships, were the two steps that were included in the analytical procedures, as stated by Anderson and Gerbing (1988). The most important reason to use PLS-SEM is that a growing number of researchers have utilized this program to examine their research models (Alam et al., 2022; Guillén et al., 2022; Kumar et al., 2020; Sitar-Taut & Mican, 2021). PLS-SEM makes it possible to evaluate the causal-predictive links between models during the process of developing and testing hypotheses (Chin et al., 2020). The variance of a target construct that can be described by predictive constructs can be increased by optimizing the correlation between causal linkages (Liu et al., 2022). Researches can anticipate that their model will have high predictive accuracy, bridging the gap between explanation and prediction (Becker et al., 2023). In addition, it provides researchers with more statistical power than factor-based structural equation modeling (Sarstedt et al., 2022). According to Hair and Alamer (2022), analysis of aggregate indicator scores can be accomplished by the utilization of a series of ordinary least squares regressions. Taking into consideration the points presented above, we have come to the conclusion that PLS is the most appropriate methodology for this research.

Furthermore, an artificial neural network (ANN) is employed on top of a partial least squares structural equation model (PLS-SEM) in order to capture linear and nonlinear interactions in a model that does not allow for compensation (Wong et al., 2020). Their ability to manage nonlinearity and multicollinearity is demonstrated by the fact that they use this strategy (Hew et al., 2018). ANN models outperform traditional statistical methods (such as MRA, SEM, and logistics) (Leong et al., 2015). This is because ANN models have a high level of accuracy compared to other statistical methods. The ability of artificial neural network (ANN) models to be taught makes them highly effective statistical models (Hew et al., 2018). PLS analysis was complemented with ANN analysis in order to rank the normalized significance of the relevant predictors. This was done since there were nonlinear interactions between the independent variables and the outcome variables. A neural network model including important predictors from a PLS-SEM input layer and a single output variable (CI) was developed by us with the assistance of IBM SPSS 26.

Common method variance (CMV)

PLS-SEM analysis would not be affected by CMV. CMV relevance in PLS analysis is disputed (Ghasemy et al., 2020). Table 5 (Appendix) shows that the inner model's VIF values are less than 3.3, making the current model CMV-free (Kock, 2015). From

study design to data collection, common method variation generated by data collection method similarity can be reduced. This study addresses procedural and statistical remedies before and after data collection. Following Rönkkö and Ylitalo (2011), the marker variable technique was utilized to check the CMV concern. PLS marker model results are compared to baseline model results. R2 changes are less than 10% in CI of 0.23%, PU of 3.7%, and PEOU of 0.49%. CMV is not a major concern in this study paradigm (Mahmud et al., 2017).

Findings

Measurement model assessment

Building the proposed model requires evaluating the outer models, according to Hair and Alamer (2022). When evaluating the outer model, factor loading, discriminate validity, composite reliability (CR), and average variance extracted (AVE) are all taken into consideration. With the exclusion of the eliminated PU1 (0.496) and RLT4 (0.638), all outer loadings are greater than Hair et al. (2019)'s proposed value of 0.708, indicating that AVE and CR have met the requirements of > 0.50 and > 0.70 (see Table 1).

Discriminant validity

According to the recommendations made by Fornell and Larcker (1981), its AVE needs to be higher than either the squared correlation of other constructs or the square root of those constructs. The fact that the diagonal has values that are higher than the values that correspond to the rows and columns in our data demonstrates that discriminant measures are present. Henseler et al. (2015) proposed the HTMT for discriminant validity evaluation. The findings can be found in Table 2. Given that the HTMT values are less

Table 1 Factor loading, CR, and AVE

Constructs	Items	Loadings	CR	AVE
Autonomy	AUT1	0.852	0.877	0.723
	AUT2	0.893		
	AUT3	0.779		
CI	CI1	0.766	0.865	0.698
	CI2	0.890		
	CI3	0.824		
Competence	COM1	0.733	0.814	0.606
	COM2	0.834		
	COM3	0.746		
PEOU	EOU1	0.812	0.873	0.713
	EOU2	0.860		
	EOU3	0.837		
PU	PU2	0.881	0.898	0.767
	PU3	0.881		
	PU4	0.829		
Relatedness	RLT1	0.749	0.853	0.676
	RLT2	0.776		
	RLT3	0.880		

Table 2 Discriminant validity

	Autonomy	CI	Competence	PEOU	PU	Relatedness
Discriminant validity (Fornell and Larcker)						
Autonomy	**0.846**					
CI	0.710	**0.775**				
Competence	0.678	0.600	**0.831**			
PEOU	0.480	0.463	0.401	**0.840**		
PU	0.516	0.383	0.500	0.526	**0.871**	
Relatedness	0.652	0.624	0.723	0.441	0.596	**0.818**
	Autonomy	**Competence**	**CI**	**PEOU**	**PU**	**Relatedness**
Discriminant validity (HTMT)						
Autonomy						
Competence	0.823					
CI	0.834	0.791				
PEOU	0.568	0.618	0.475			
PU	0.608	0.490	0.610	0.613		
Relatedness	0.805	0.843	0.817	0.527	0.729	

Values on the diagonal (bold) represent the square root of the AVE, while the off-diagonals are correlations

Table 3 Coefficient of determination

Endogenous variables	R Square
CI	0.51
PEOU	0.29
PU	0.49

than the HTMT.90 value of 0.90, Gold et al. (2001) state that discriminant validity is not a concern.

Structural model assessment

Multicollinearity test (VIF)

Before assessing the inner model, check for collinearity to avoid biased regression findings (Hair et al., 2019). The partial regression scores of the latent variables on the predictor constructs yield a variance inflation factor (VIF). Multicollinearity indicates redundant information from associated predictors. The VIF found value. A small VIF means variables are not closely connected. VIF values below 3.3 are excellent (Kock, 2015). In the study model, all VIFs are below 3.3 (see Table 5). It was concluded that the research model had no multicollinearity concerns.

Coefficient of determination (R^2)

The impact level for R^2 is zero to one. Cohen (1988) considered R^2 values of 0.26, 0.13, and 0.02 considerable, moderately strong, and weak for endogenous variables. Table 3 reveals high predictive accuracy for CI, PEOU, and PU in this investigation. Relatedness,

Table 4 Predictive power assessment using $PLS_{predict}$

Items	$Q_{predict}^2$	PLS-SEM_RMSE	Linear model_ RMSE
CI1	0.240	1.148	1.260
CI2	0.397	1.064	1.110
CI3	0.411	1.184	1.233
PEOU1	0.077	1.195	1.172
PEOU2	0.141	1.334	1.147
PEOU3	0.220	1.219	1.191
PU2	0.233	0.879	0.962
PU3	0.299	1.127	1.020
PU4	0.278	1.234	1.235

Table 5 Path coefficient and hypothesis test result

No.	Relationship	Std. beta	Std error	t-value	*P* value	f^2	Effect size	VIF	Decision
H1	Autonomy → CI	0.577	0.099	5.017	0.000	0.434	Large	1.506	Supported
H2	Autonomy → PEOU	0.238	0.127	1.645	0.032	0.025	Small	2.456	Supported
H3	Autonomy → PU	0.220	0.101	1.859	0.014	0.026	Small	2.536	Supported
H4	Competence → PEOU	0.196	0.138	1.257	0.085	0.004	No effect	2.306	Not supported
H5	Competence → PU	(0.202)	0.092	1.848	0.015	0.031	Small	2.366	Supported
H6	PEOU → CI	0.026	0.087	0.231	0.387	0.016	No effect	1.526	Not supported
H7	PEOU → PU	0.325	0.129	2.224	0.000	0.134	Medium	1.396	Supported
H8	PU → CI	0.195	0.092	1.797	0.019	0.034	Small	1.616	Supported
H9	Relatedness → PEOU	0.173	0.113	1.322	0.074	0.004	No effect	1.966	Not supported
H10	Relatedness → PU	0.445	0.111	3.496	0.000	0.174	Large	2.006	Supported

Table 6 Testing model fit

Parameter	Saturated model	Estimated model
SRMR	0.061	0.075
NFI	0.902	0.927

competence, and autonomy describe 29% of PEOU variation. Also, they affect PU variation by 49%. Lastly, 51% of the variation in CI may be explained by PEOU and PU.

PLS predictive power

Using the model's predictive power for endogenous constructs (PEOU, PU, CI), ten folds and ten iterations of $PLS_{predict}$ were used (Sharma et al., 2022). We first examined if $Q^2_{predict}$ was above zero, suggesting that the PLS route model exceeded the training data indicator means. PLS-SEM estimates were compared to a linear benchmark model's RMSE (Hair et al., 2019). Table 4 shows $PLS_{predict}$ results. All endogenous construct indicators had positive $Q^2_{predict}$ values. Compared to the linear model, PLS-SEM had lower

RMSE values overall. Overall, the PLS path model predicts the endogenous constructs of interest strongly because all item differences (PLS-LM) are smaller (Shmueli et al., 2019).

Path coefficient

An estimation of the path coefficients for the proposed research model was carried out using bootstrap t-values with 5,000 resamples (Henseler et al., 2009). Path coefficients range from −1 to +1. A strong positive association is produced when the value is around 1; −1 is negative. Thus, the path coefficients' least significant level should be 0.05. At 10%, 5%, and 1% significant levels, critical values of 1.645, 1.96, and 2.33 are assigned to the one-tailed test (Ramayah et al., 2014). Out of the ten hypotheses evaluated in the study, Table 6 reveals that three hypotheses (i.e., H4, H6, and H9) were not supported by the data. The effect size was estimated using Cohen's method (Cohen, 1988). Effect sizes of 0.02, 0.15, and 0.35 are small, medium, and substantial. Relatedness and autonomy greatly affect CI and PU. PEOU moderately affects PU, while other factors are negligible. PEOU did not influence CI (see Table 5).

Model fit and goodness of fit (GoF)

The model's accuracy was initially assessed by fitting it to two adjustment variables: the normed fit index (NFI) and the standardized root mean square residual. SRMR values less than 0.08 (calculated by subtracting the observed correlation matrix from the model-implied correlation matrix) indicate a well-fit model (Hu & Bentler, 1998). In the PLS-SEM approach, Henseler et al. (2014) recommended the SRMR as a goodness-of-fit metric to avoid model misspecification. The normed fit index is the second model fitness metric. Ramayah et al. (2017) present a second model fitness metric that compares the proposed model's chi-square value to a meaningful standard. NFI values above 0.90 usually suggest a satisfactory match (Bentler & Bonett, 1980). The saturated (measurement) model had no free routes; therefore, calculating the structural model and fitting it to it yielded comparable results. Table 6 shows that the data matched the model well, with an

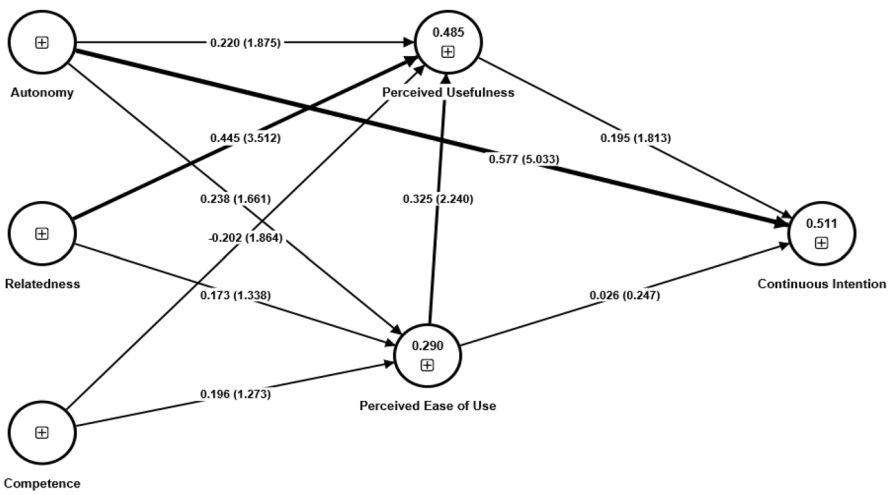


Fig. 2 The research model

SRMR of 0.075 (< 0.08) and an NFI of 0.927 (> 0.90). Figure 2 displays the test results in the final research model. Using the square root of the mean of R^2 multiplied by the mean of AVE, Tenenhaus et al. (2005) presented a model called the GoF. Therefore, the number 0.360 is appropriate, as stated by Wetzels et al. (2009). If the proposed model has a GoF value of 0.548, this indicates that the model has been adjusted appropriately. The results of the tests are displayed in the final research model, which can be found in Fig. 2.

ANN analysis

The application of a multi-analytical technique was accomplished by combining multiple PLS-SEM and ANN. Due to the nonlinear interactions that exist between the independent factors and the outcome variables, we conducted an additional analysis with multilayer perceptron (MLP) and artificial neural networks (ANN) in order to rank the normalized importance of the predictors. This was done in order to confirm the findings of the PLS-SEM. In light of the fact that the PLS-SEM can only be used to investigate the validity of linear hypotheses and cannot take into consideration the nonlinearity of correlations, the utilization of a technique that operates in two stages would be complimentary. The artificial neural network (ANN) is unable to test hypotheses due to its "black box" function, despite the fact that it is capable of recognizing nonlinear correlations (Hew & Syed Abdul Kadir, 2016; Karkonasasi et al., 2018). Additionally, the ANN is resistant to noise, outliers, and limited sample sizes, among other factors. It is also capable of supporting non-compensatory models, which are models in which a decrease in one element does not always necessitate an increase in another factor (Lee et al., 2023). For the purpose of carrying out the ANN analysis, we utilized IBM SPSS 26. In the succeeding phase, we used the important factors of the latent construct scores that were produced using PLS-SEM as input neurons for the ANN model. This was done in the same manner as Leong et al. (2020) and Abdullah et al. (2022). A feed-forward–backward-propagation (FFBP) approach, in which inputs are given in a forward channel, and the estimated errors flow in a backward direction, can be utilized to train the system to predict the outcomes of the study (Liébana-Cabanillas et al., 2017). This algorithm allows for the system to be taught to predict findings from the study. A multilayer perceptron with hyperbolic tangent activation functions was selected for the hidden layer and the identity function output. This decision was made in light of the fact that the

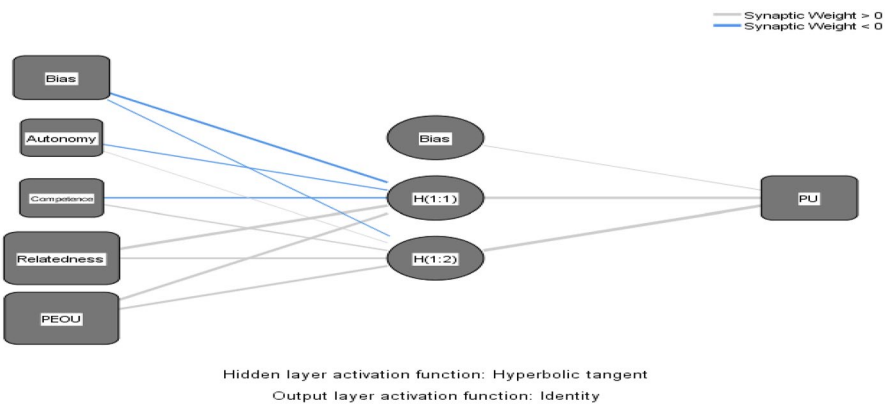


Fig. 3 The structure of the ANN model A

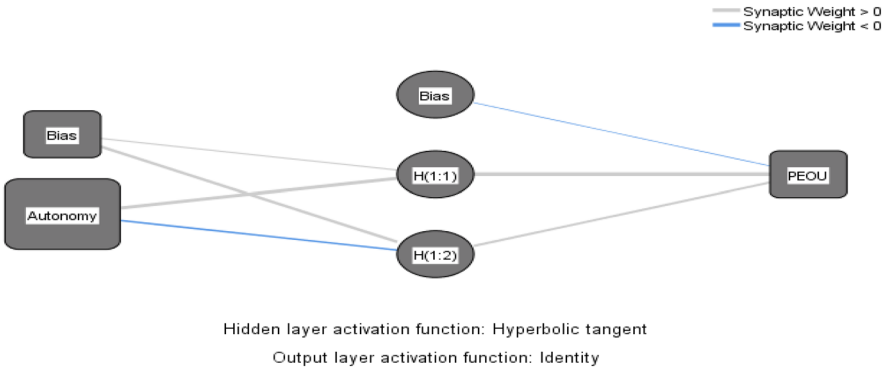


Fig. 4 The structure of the ANN model B

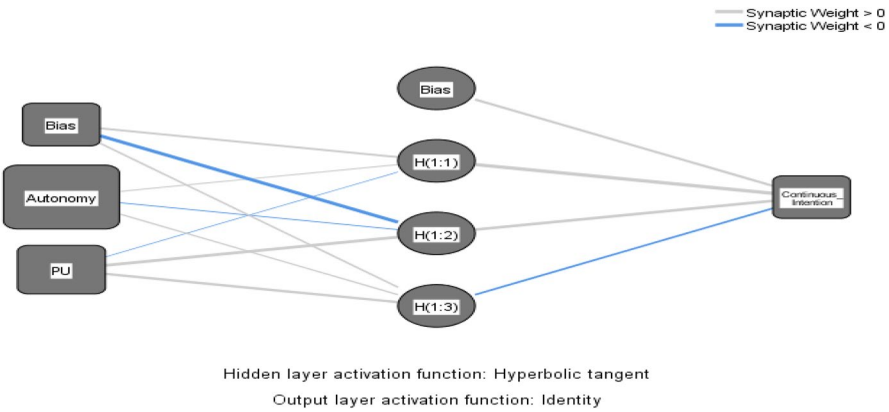


Fig. 5 The structure of the ANN model C

Table 7 RMSE values

Neural network	Model A		Model B		Model C	
	Input: autonomy, relatedness, competence, PEOU		Input: autonomy		Input: PU, autonomy	
	Output: PU		Output: PEOU		Output: CI	
	Training RMSE	Testing RMSE	Training RMSE	Testing RMSE	Training RMSE	Testing RMSE
NN 1	0.523	0.526	0.585	1.195	0.464	0.456
NN 2	0.659	0.457	0.632	0.654	0.492	0.259
NN 3	0.511	0.729	0.636	0.842	0.44	0.688
NN 4	0.532	0.491	0.603	0.475	0.475	0.468
NN 5	0.585	0.679	0.605	0.917	0.467	0.286
NN 6	0.52	0.596	0.618	0.51	0.523	0.637
NN 7	0.571	0.259	0.591	0.72	0.431	0.591
NN 8	0.547	0.678	0.622	0.577	0.47	0.214
NN 9	0.537	0.183	0.629	0.245	0.499	0.325
ANN 10	0.536	0.493	0.599	0.437	0.507	0.233
Mean	0.552	0.509	0.612	0.657	0.477	0.416
SD	0.033	0.167	0.007	0.263	0.018	0.166

SSE = Sum of square error, RMSE = Root mean square of errors, SD = Standard Deviation

Table 8 Average weights of the input and hidden neurons in the ten-fold ANN (A)

Predictors	Predicted hidden layer 1		Output layer
	H(1:1)	H(1:2)	PU
Input layer			
(Bias)	−0.423	−0.319	
Autonomy	−0.324	0.172	
Competence	−0.364	0.335	
Relatedness	0.759	0.395	
PEOU	0.507	0.490	
Relatedness	−0.423	−0.319	
Hidden layer 1			
(Bias)			0.174
H(1:1)			0.541
H(1:2)			0.761

Table 9 Average weights of the input and hidden neurons in the ten-fold ANN (B)

Predictors	Predicted Hidden Layer 1		Output Layer
	H(1:1)	H(1:2)	PEOU
Input layer			
(Bias)	0.074	0.489	
Autonomy	0.610	−0.249	
Hidden layer 1			
(Bias)			−0.039
H(1:1)			0.862
H(1:2)			0.368

output is continually scaled. On the other hand, we only employ the sigmoid activation function in situations when the output layer is reliant on a binary or multicast categorical function (Taufiq-Hail et al., 2021) (Figs. 3, 4, 5).

By following the recommendations of Leong et al. (2020), we used 90% of the samples for the training phase, and the remaining 10% were used for the testing procedure. For the purpose of avoiding the possibility of overfitting, we employed a ten-fold cross-validating strategy and computed the root mean square of errors (RMSE) (Wong et al., 2020). According to Taufiq-Hail et al. (2021), several learning rounds have the potential to enhance the accuracy of predictions while simultaneously lowering the number of errors that occur.

Table 7 demonstrates that the regression mean square error (RMSE) values of the training and testing techniques are relatively low. The values for models A, B, and C are 0.552, 0.612, and 0.477, respectively. That being said, we are able to demonstrate that the model fits the data extraordinarily well while maintaining a high level of forecast accuracy. Our findings indicate that the ANN model is capable of accurately predicting PU with a 62.7% accuracy, predicts PEOU with an accuracy of 24.4%, and predicts CI with an accuracy of 58.8%. We calculated the R2 of the ANN model using a methodology that is comparable to the one that was utilized by Hew and Syed

Table 10 Average weights of the input and hidden neurons in the ten-fold ANN (C)

Predictors	Predicted hidden layer 1		H(1:3)	Output layer
	H(1:1)	H(1:2)		Continuous intention
Input layer				
(Bias)	0.432	−1.961	0.398	
Autonomy	0.364	−0.124	0.388	
PU	−0.015	1.490	0.463	
Hidden layer 1				
(Bias)				0.457
H(1:1)				2.013
H(1:2)				1.476
				−0.409

Table 11 Sensitivity analysis

Model A (output: PU)			Model B (output: PEOU)			Model C (output: CI)		
Variable	Average importance	Normalized importance	Variable	Average importance	Normalized importance	Variable	Average importance	Normalized importance
Autonomy	0.536	69%	Autonomy	1.00	100%	PEOU	0.62	64%
Competence	0.360	48%	–	–		PU	1.00	100%
Relatedness	0.834	100%	–	–		–	–	–
PEOU	0.836	100%	–	–	–	–	–	–

Abdul Kadir (2016) and Leong et al. (2020). After the H (1:2) hidden neuron, which is an inhibited cryptic neuron, the H (1:2) hidden neuron is the most significant for PU as the output layer, as shown in Table 8 (Model A). As can be shown in Table 9 (Model B), the H (1:1) hidden neuron is the most significant hidden neuron for PEOU in the output layer. This is followed by the H (1:2) hidden neuron, which is an indirect cryptic neuron. The hidden neuron of H (1:1) is the one that contributes the most to CI as the output layer, as seen in Table 10 (Model C). This is followed by H (1:2), which is an inhibited cryptic neuron. H (1:3), on the other hand, is an example of a cryptic neuron that has been repressed by the process of suppression. Considering that every single non-zero synaptic weight is connected to at least one hidden neuron, it can be concluded that every single ANN model utilized in this inquiry possesses an adequate amount of predictive significance.

By dividing the relative value of these neurons by the highest importance and expressing the result as a percentage, we were able to compute the normalized relevance of these neurons through the use of sensitivity analysis (see Table 11) (Abdullah et al., 2022). According to the findings, the most significant predictors for PU (Model A) are relatedness and PEOU (100%), followed by authority (69%) and competence (48%). Similar to the previous example, the most significant predictor of PEOU (Model B) is the degree of autonomy (100%). One hundred percent of the time, PU is the most significant predictor for CI (Model C).

Table 12 Comparison between PLS-SEM and ANN results

PLS path	Path coefficient	ANN results: normalized relative importance	Ranking (PLS-SEM_ [based on path coefficient]	Ranking (ANN) [based on normalized relative importance]	Remark
Model A (output: PU)	–	–	–	–	–
PEOU → PU	0.325	100%	2	2	Match
Relatedness → PU	0.445	100%	1	1	Match
Autonomy → PU	0.220	69%	3	3	Match
Competence → PU	-0.202	48%	4	4	Match
Model B (output: PEOU)	–	–	–	–	–
Autonomy → PEOU	0.238	100%	1	1	Match
Model C (output: CI)	–	–	–	–	–
PU → CI	0.195	64%	2	2	Match
Autonomy → CI	0.577	100%	1	1	Match

When it came time to compare the outcomes of PLS-SEM and ANN, the relative rankings of each method were done. The normalized relative importance in ANN was compared to the path coefficients in PLS-SEM, which permitted this to be accomplished. Both of the evaluations are in line with the three ANN models (A, B, and C) that are presented in Table 12.

Discussion

This study's primary objective was to investigate the factors that influence the intention of Egyptian university students to use the metaverse as a learning platform on a continual basis. An integration of two theoretical models, namely TAM and SDT, has been successful in achieving this research. The results that were acquired from the empirical survey are the ones that are presented in the discussion section. We discovered evidence that supports the seven additional hypotheses that were provided for the model. There is a favorable relationship between autonomy and relatedness in both PU and PEOU. As a result, autonomy and relatedness are major determinants of the outcome. Cortez et al. (2024), Wang and Shin (2022), and Racero et al. (2020) came to the same conclusion, which is consistent with our outcome. In light of this discovery, it can be deduced that the metaverse, when utilized as a learning platform, becomes advantageous and simple to employ if it possesses the qualities of relatedness and autonomy. On the other hand, competency does not have any influence on PU and PEOU. Therefore, expertise is not a significant factor in predicting this. As a result of the fact that Jeno et al. (2017) and Liaw and Huang (2015) discovered that competence was a significant component in PU and PEOU, this finding is in direct opposition to their investigations. Possible explanations for this disparity include the fact that Egypt and other countries have different cultural norms and traditions. Teachers, parents, or private tutors are the primary sources

of support for kids in Egypt, for instance. As a consequence of this, they might not feel comfortable utilizing the metaverse as a platform for information acquisition. Likewise, autonomy and PU are seen as strong predictors. If students have the sense that they are valuable and independent, they are more likely to continue using the metaverse as a learning tool. These results are in line with the findings that were produced by Al-Adwan and Al-Debei (2024), Kye et al. (2021), and Racero et al. (2020). The unanticipated finding of this study, on the other hand, demonstrates that PEOU was not a major predictor in their ability to influence the continuing intention to use the metaverse as a learning platform. Wiangkham and Vongvit (2024) came to the same conclusion, which is consistent with this outcome. This is the first time that the majority of pupils have been required to utilize the metaverse; therefore, they do not have any prior understanding or experience with it.

Further, the dual analytic strategy, which consists of PLS-SEM and ANN, was utilized in order to improve and offer new insights into the findings. A comparison of the strengths of both methodologies was made in order to evaluate the linear (PLS-SEM) and nonlinear (ANN) interactions. The findings were further magnified to address the study questions by the causal links that were verified by PLS-SEM and the predictive power of the ANN. A two-staged strategy of PLS-SEM with a nonlinear, no compensatory neural network model was utilized in this research project, which resulted in the effective verification of the significant factors of CI in the utilization of metaverse in learning. One hundred percent of the normalized importance is assigned to the variables of relatedness, PEOU, autonomy, and PU. These are the most important predictors. CI's use of the metaverse may be predicted with a level of accuracy of 58.8% by the ANN model. Not only that, but the ranking comparison between the findings that were given by ANN and those that were provided by PLS-SEM reveals that the three ANN models (A, B, and C) are consistent throughout both assessments (see Table 12).

Theoretical and practical implications

The work detailed in this paper is responsible for a great deal of theoretical progress. In order to enhance the current body of knowledge, this study constructed a novel hybrid model that incorporates TAM and SDT. Very few research have addressed the topic of using the metaverse as a learning platform; however, this model was utilized to forecast continual intention to do so in Egypt. Secondly, this study presents a number of important conclusions, one of which is that, when it comes to the ongoing purpose to learn in the metaverse, autonomy was found to be more dependable than PU. Thirdly, with regard to the elements influencing PU and PEOU, this research presents a number of important conclusions. The two most important results are relatedness and autonomy. When it comes to PU, autonomy is far more important than relatedness. On the other hand, compared to autonomy, relatedness significantly impacts PEOU. A new theoretical contribution was also made to the literature by the study. The reason is that the linear compensatory model is not perfect; after all, an increase in one predictor might make up for a decrease in another. In order to compensate for the linear model's limitations, the present work introduces a non-compensatory ANN model (Taufiq-Hail et al., 2021). Also, this study builds on prior work by creating a PLS marker variable approach to reduce the impact of common method variance (CMV) bias on the PLS structural route.

The results provide strong evidence that this study also builds upon earlier work in the field. It is clear from the results that.

We can propose a set of practical consequences based on these important findings. To begin, in order for the metaverse to serve as a learning platform, it must incorporate methods that facilitate relatedness features, including online discussion forums, chat rooms, instant messaging, and message boards. These concepts can make the metaverse more practical and easier to use, which will enhance the likelihood that people will continue to utilize it for learning. Second, students' autonomy is crucial, and it may be improved by providing them with tools that let them set their own learning speed and cultivate a sense of freedom and choice. Lastly, PU plays a crucial role in enhancing the long-term goal of utilizing the metaverse for educational purposes. Improving it with time-saving, learning-enhancing, and information-accessibility-enhancing features is possible. Additionally, the dual-stage hybrid SEM-ANN approach creates a more accurate interpretation of the interrelationships among variables in the context of education, at least from the practitioners' insights. The nonlinear approach broadened the practical perspective by predicting the variables that have an effect on the intention to utilize the metaverse as a learning platform on a continuing basis. Additionally, the restricted exploration of typical virtual learning patterns based on linear and nonlinear correlations in education is still an open question for academics and practitioners.

Conclusion

Through the conceptualization and empirical testing of a model that blends TAM and SDT theories, the current research contributes to the expansion of the existing body of knowledge. University students were given a questionnaire to fill out as part of a survey that was being conducted. During the analysis of the data, the PLS-SEM approach was utilized. When it came to the decision of whether or not students would continue to use the metaverse for educational purposes in the future, autonomy, and PU were significant indications. On the other hand, PEOU was not a significant factor in determining continuous intention. In addition, the research showed that autonomy and relatedness are believed to be important predictors of both PU and PEOU, whereas competence was shown to be an inconsequential predictor. Furthermore, according to the ANN model, relatedness, PEOU, autonomy, and PU are the most significant predictors, with a normalized value of 100%. The results that are obtained through the use of the PLS-SEM and ANN modes are identical.

Limitation and future work

This piece of work has a few shortcomings that need to be discussed and recognized. As a result of time constraints, the current study is cross-sectional, which means that it quantifies intentions at a certain point in time. The accumulation of experience causes individuals to develop new perspectives during the course of their lives. To this end, the longitudinal method is the most appropriate option to use. Only one private university was represented among the participants in the survey; there were no representatives from public universities. Therefore, it is not possible to generalize the findings of this study to other higher education institutions in Egypt. As a consequence of this, it is possible that future research will involve students from both public and private universities

in order to contrast two distinct scenarios and generalize the findings. If various theories of technology acceptance are incorporated into future studies, it will be possible to gain a better understanding of the areas of interest that students have as well as the factors that influence their adoption of new technologies. It is possible that future research will take into account more key aspects, such as cognitive capacity, social relationships, and concerns about safety. The provision of connectivity can be of assistance in the development of virtual learning systems, particularly in light of the fact that a dependable and stable internet connection is quite important for metaverse technologies. The functionality of electronic devices may be negatively impacted if enough infrastructural support is not provided enough. The conclusion that can be drawn from this is that internet services must be of the best possible quality. When it comes to teaching, qualitative methodologies should be utilized in order to gain a deeper comprehension of the attitudes and behaviors of younger students. Take, for instance, the possibility of gaining more information through the use of participant interviews or focus groups.

Abbreviations

ANN	Artificial neural network
AR	Augmented reality
AVE	Average variance extracted
BCI	Brain–computer interfaces
CI	Continuous intention
CR	Composite reliability
CMV	Common method variance
FFBP	Feedforward-backward-propagation
GPUs	Graphics processing units
GoF	Goodness of fit
HEIs	Higher education institutions
HTMT	Heterotrait–Monotrait ratio of correlations
MLP	Multilayer perceptron
MR	Mixed reality
NFI	Normed Fit Index
PEOU	Perceived ease of use
PLS-SEM	Partial least squares structural equation modeling
PU	Perceived usefulness
RMSE	Root mean square of errors
SDT	Self-determination theory
SRMR	Standardized root mean square residual
TAM	Technology acceptance model
3D	Three-dimensional
VR	Virtual reality
VIF	Variance inflation factor
XR	Extended reality

Acknowledgements
The authors would like to greatly thank the Prince of songkla University (PSU) for providing any funding and facilities for this research. Also, the authors are grateful to the anonymous referees of the journal for their extremely useful suggestions to improve the quality of the paper.

Author contributions
Reham Adel Ali: Conceptualization, methodology, data collection, writing original draft. Mohamed Soliman Mohamed Soliman*: Corresponding author, writing, review and editing original draft, and formal analysis. Muhammad Roflee Weahama: Project administration, supervision, and review. Muhammadafeefee Assalihee: Visualization, review, and editing. Imran Muhmud: Provided scholarly insights and guidance, formal analysis, review, and editing. In addition, all authors read and approved the final manuscript.

Funding
This research is under the funding of Prince of songkla University (PSU), grant number ISL6602068S.

Availability of data and materials
Data will be made available on a reasonable request.

Declarations

Competing interests
The authors declare that they have no competing interests.

Received: 19 June 2023 Accepted: 20 October 2024
Published online: 13 January 2025

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